PRD QR Manager Settlement Report

SANITIZED TEST COPY - CONFIDENTIAL VALUES REPLACED WITH SYNTHETIC DUMMY DATA

Field	Dummy value
Product Name	QR Manager Settlement Report
Epic Link	QRM-4821: [QR Manager] Improvement QR Manager Settlement Report - Ready for SIT
Project No	482/DEMO/QR/2.03/IX/26
Document Status	IN PROGRESS
Product Manager	@Nadia Pratama
Lead Product Manager	@Raka Wijaya
Tech Lead	@Maya Kurniawan
Engineering Manager	@Bima Santoso

Document Approval

Created by	Name	Title	Project Role	Signature
	Nadia Pratama	Product Manager	Product Manager	

Reviewed by	Name	Title	Project Role	Signature
	Maya Kurniawan	Tech Lead	Tech Lead	

Approved by	Name	Title	Project Role	Signature
	Raka Wijaya	Lead Product Manager	Product Manager	
	Bima Santoso	Engineering Manager	Engineering Manager	

All names, handles, identifiers and signatures in this copy are synthetic and are intended only for QATOOL testing.

Background

The QR Manager application serves as the core platform for managing and executing QRIS settlement activities. Currently, the Settlement Report module faces operational bottlenecks due to heavy data loading, manual extraction workflows, and rigid UI constraints. The existing system lacks dynamic pagination and features a limited Advanced Search function that only supports a single combined data view. These limitations prevent the Reconciliation and Operations teams from efficiently filtering specific transaction states, reviewing pending tasks, and exporting critical financial data on demand.

Additionally, infrastructure timeouts during settlement releases to the Core Banking system via Middleware pose significant operational risks, as the application currently lacks an isolated, automated recovery mechanism. To resolve these challenges, QR Manager requires an upgrade to optimize report rendering, expand search filtering capabilities, automate daily data pipelines for downstream reconciliation systems, and deploy a resilient asynchronous timeout engine to secure end-to-end settlement operations.

Objective

- Improve report rendering through database tuning, updated page limits, and a new release timestamp column
- Expand filtering options within the advanced search criteria and ensure the export button is always functional
- Automate the daily generation and transfer of settlement report files for reconciliation purposes
- Implement an isolated queue and automated check-status retry loop to handle timed-out transactions without blocking the main workflow

Relevant Documents

Reference	Status
QRM-4821-650: Penambahan Switching Atlas / ATMZ untuk layanan QRIS DemoBank	Ready for Deployment

Requirements

1. Set Default View, Implement Pagination and Add New Column on Transaction Data

Scenario	Acceptance Criteria	Jira Link
Default Table in Settlement Report Menu	**GIVEN** User is on the QR Manager Dashboard **WHEN** User opens the Settlement Report menu **THEN** the system must fulfill: - Automatically load and show transactions with the status Unsettled or Unreleased - Ensure to load all available data from all dates	QRM-17394: Settlement Report - Default View
Row Display and Pagination Limit	**GIVEN** User is viewing the default Settlement Report table **WHEN** the data is fully rendered on screen **THEN** the system must fulfill: - Display a maximum of 10 rows of transaction per page - Display the navigation control buttons as Prev 10 and Next 10 (current condition: Prev 100 and Next 100)	QRM-17394: Pagination update

2. Make the Export Button Always Active

Scenario	Acceptance Criteria	Jira Link
Add Release Date and Time Column	**GIVEN** QR Manager User is viewing the transaction data tables within the Settlement Report menu **WHEN** a transaction status has successfully achieved Settled or Released **THEN** the system must fulfill: - Add and display new data column for RELEASE DATE AND TIME to the table - Include these timestamps inside the report file **Reference:** QRM-17394 - Default View / Pagination / Release Timestamp	QRM-17394: Release timestamp
Activate the Export Button	**GIVEN** QR Manager User is on the Settlement Report menu with data rendered **WHEN** user moves the cursor over or clicks Export button **THEN** the system must fulfill: - Maintain the Export to TXT button status as active and enabled under all data states - Trigger the file generation or download process immediately upon clicking without any validation lockouts **Current Export Button:** Export is unavailable for date ranges greater than 1 day.	QRM-17449: Always Active Export

3. Table Tuning to Fix Report Slowness

Scenario	Acceptance Criteria	Jira Link
Table Tuning	**GIVEN** QR Manager User is on the QR Manager Dashboard **WHEN** QR Manager User opens the Settlement Report menu **THEN** the system must fulfill: - Perform Table Tuning to optimize data retrieval process - Load and display the transaction data grid quickly **Current condition in production when opening Settlement Report menu (cost 28 seconds):**	QRM-17451: Table Tuning

4. Auto-Generate Daily Settlement Report

Scenario	Acceptance Criteria	Jira Link
Auto-Generate Daily Settlement Report	**GIVEN** all transactions for the D-1 (previous day from 00.00-23.59) have concluded **WHEN** the system clock reaches 03.00 AM / off-peak hours **THEN** the system must fulfill: - Create and implement new automated scheduler to trigger the daily report generation - Extract all transaction data from 00.00 to 23.59 on D-1 (previous day) including both H+1 and On Hold transactions - Ensure the process runs automatically without normal intervention and without creating temporary files (.tmp) during extraction - Use SETTLEMENT_QRMANAGER_DDMMYYYY.txt as file name format (Date using Transaction Date) - Generate the final report file in .txt format using a tilde ~ as the data delimiter - Save the final .txt file into a designated directory path on QR Manager server so it is ready to be fetched by Reconciliation team. - Report format to be generated as shown below.	QRM-17479: Auto-Generate Daily Settlement Report

Synthetic report-format sample

Column Name Report	Source Table Name	Source Column Name	Format	Sample Data
TRX_DATE	v_qr_release_settlement	TRX_DATE	DD-MMM-YYYY	14-Oct-26
SETTLEMENT_DATE	v_qr_settlement_group	SETTLEMENT_TIME	DD-MMM-YYYY HH:mm:ss	14-Oct-26 03:04:12
RETRIEVAL_REF_NUM	v_qr_release_settlement	RETRIEVAL_REF_NUM	VARCHAR	QRM26091400127
MERCHANT_ID	v_qr_release_settlement	MERCHANT_ID	VARCHAR	DMBK000045
PAN	v_qr_release_settlement	PAN	Decimal	400012******9012
MERCHANT_NAME	v_qr_release_settlement	MERCHANT_NAME	VARCHAR	Demo Corner Merchant
MERCHANT_CRITERIA	v_qr_release_settlement	MERCHANT_CRITERIA	VARCHAR	UMI
MERCHANT_SETTLEMENT_TYPE	v_qr_release_settlement	MERCHANT_SETTLEMENT_TYPE	VARCHAR	H+1
TO_ACCOUNT	v_qr_release_settlement	TO_ACCOUNT	VARCHAR	8899001122
AMOUNT	v_qr_release_settlement	AMOUNT	Decimal 2 digits	125000
CHARGE_AMOUNT	v_qr_release_settlement	CHARGE_AMOUNT	Decimal 2 digits	1250
AMOUNT_SETTLEMENT	v_qr_release_settlement	AMOUNT_SETTLEMENT	Decimal 2 digits	123750
FT_NUMBER_SETTLEMENT	v_qr_release_settlement	FT_NUMBER_SETTLEMENT	VARCHAR	FTDEMO014922
SWITCHING_NAME	v_qr_release_settlement	SWITCHING_NAME	VARCHAR	ALTO-DEMO
ID_GROUP	v_qr_release_settlement	ID_GROUP	VARCHAR	GRP2609140008
SETTLEMENT_STATUS	v_qr_settlement_group	SETTLEMENT_STATUS	VARCHAR	SUCCESS
SETTLEMENT_GROUP_ID	v_qr_settlement_group	SETTLEMENT_GROUP_ID	VARCHAR	SG260914000012

Connectivity note (sanitized): Scope connectivity QR Manager only: QR Manager <-> SFTP <-> ReconcileWorks. Demo production endpoint: 192.0.2.45:8080. Path: /opt/demo/recon. All values are synthetic.

5. Redefine Timeout Transaction Logic via Check Status

Scenario	Acceptance Criteria	Jira Link
Timeout Catching & Delay Buffer	**WHEN** QR Manager proceeds Release Settlement to Core Banking via Middleware and experiences Timeout (TTTP 408 or 504 or responseCode 68 - Timeout) **THEN** the system must fulfill: - Mark the transaction as PENDING and route the record into a dedicated CHECK STATUS queue. - Trigger a scheduler to manage this queue, preventing any resource blocking on the main transaction flow. - Enforce a strict 1 hour delay from the original timeout timestamp before allowing the scheduler to process the record (giving the Core Banking system a cool-down period).	QRM-17521: Timeout Transaction Logic

Synthetic infrastructure note

/opt/demo/recon - this path is mounted to a synthetic downstream server for test purposes.

Status Verification and Conditional Retry

Scenario	Acceptance Criteria	Jira Link
Status Verification and Conditional Retry	**GIVEN** the interval time has passed and the isolated scheduler picks up the PENDING transaction **WHEN** the system executes the status verification **THEN** the system must fulfill: - Call Check Status to Core Banking via Middleware to verify the actual state of the transaction. - **IF** check status response from Core Banking is Success (Y.ISO.RESPONSE:1:1="00" and Y.MSG.RES:1:1="{{FT Number returned}}"), update the DB status to SUCCESS and save the FT NUMBER. - **IF** FAILED (Y.ISO.RESPONSE:1:1="01") or NOT FOUND, then trigger retry Release Settlement to Middleware. The maximum number of settlement release retries is 3. - **IF** after Release Settlement 1 and/or 2 retry attempts the transaction still Timeout, perform Check Status again. - **IF** the status remains unresolved (FAILED) after 3rd retry attempt, permanently mark the transaction status as FAILED. Notes: Retry for Settlement Release is only performed when the status check response is FAILED (transaction not completed) and the retry attempt has not reached 3 times.	QRM-17521: Check Status Retry

Synthetic OFS request via middleware

```
"!OPERATION"      : "ENQUIRY.SELECT",
"!TRANSACTION_ID" : "E.CHN.VERIFY.FT.DEMO",
"UMG.UNIQUE.ID:EQ": "UMG-DEMO-001998772",
"QUEUE.NAME"      : "SETTLEMENT-CHECK"
```

Synthetic response from Core Banking via middleware

```
{
  "responseCode": "00",
  "responseMessage": "Success!",
  "data": {
    "RESPONSE.TRX::": "01",
    "Y.MSG.RES:1:1": "FTDEMO014922",
    "rawResponse": "...synthetic response...",
    "rawRequest": "ENQUIRY.SELECT,,UMG-DEMO-001998772,E.CHN.VERIFY.FT.DEMO"
  }
}
```

Response Mapping

```
Y.ISO.RESPONSE:1:1="00" AND
Y.MSG.RES:1:1="FTDEMO014922"

Funds Transfer (FT)
Success (Transaction successfully processed in core)

Y.ISO.RESPONSE:1:1="01" AND
Y.MSG.RES:1:1="Transaction Failed!"

Transaction Not Found / Failed
(FT is not recorded)
```

Core Banking Response Values	Transaction Meaning / Mapping
Y.ISO.RESPONSE:1:1="00" AND Y.MSG.RES:1:1="FTDEMO014922"	Funds Transfer (FT) Success (Transaction successfully processed in core)
Y.ISO.RESPONSE:1:1="01" AND Y.MSG.RES:1:1="Transaction Failed!"	Transaction Not Found / Failed (FT is not recorded)

6. Add Filter in Advanced Search

Synthetic example values used in supporting screenshots / tables

```
UE.ID:EQ=QRDEMO7589650707132337015301535910 2
correlationId: QRM-DEMO-90acc398738142d79995bccafd9433ca
Y.ISO.RESPONSE:1:1="00"
Y.MSG.RES:1:1="FTDEMO014922"
```

6. Add Filter in Advanced Search

Scenario	Acceptance Criteria	Jira Link
Add New Filter Options	**GIVEN** QR Manager User open Settlement Report menu **WHEN** User click Advance Search filter dropdown **THEN** the system must fulfill the following: - Add a new dropdown field named Settlement Status: placed directly below the existing Settlement Type dropdown: Settled, Unsettled, and Settled & Unsettled - Process and render the data according to the selected filter within the chosen time period: • IF the User pick Settled filter, then display all of the settled transaction • IF the User pick Unsettled filter, then display all of the unsettled transaction • IF the User pick Settled & Unsettled, then display all of the settled and unsettled transaction	QRM-19347: Add New Advance Search Filter

Synthetic filter values: Settlement Type = H+1 & Hold; Settlement Status = Settled / Unsettled / Settled & Unsettled.

7. Tick All QR Transaction Release Settlement

Scenario	Acceptance Criteria	Jira Link
Tick All QR Transaction Release Settlement	**GIVEN** the QR Manager user is on the Settlement Report page **WHEN** the user clicks the Release action button (Settlement Type: HOLD) **THEN** the system must fulfill the following: - Display a Select All checkbox on the page near the transaction list table - Select all transaction rows in the table when the user checks the Select All box - Unselect all transaction rows in the table when the user unchecks the Select All box **Notes:** Select All only applies to transactions on the current active page (up to 10 rows per page) and will not select items on other pages.	QRM-19411: Select All Release Settlement

Synthetic UI example

QR Channel	FT Number	Status	Release Transaction
QRDEMO001	FTDEMO014922	PENDING	■
QRDEMO002	FTDEMO014923	PENDING	■
QRDEMO003	FTDEMO014924	SUCCESS	■
		Select All	■

Example behavior: checking Select All selects only rows visible on the active page (maximum 10 rows), while rows on other pages remain unchanged.

Rollout Plan

Stage	Status / Notes
SIT	
Performance Test / Load Test	If needed
UAT	
SAT	
LIVE	

Sanitization Notes

- All personal names and user handles were replaced with synthetic names.
- Project, Jira/issue, retrieval, FT, transaction, merchant, account, correlation, unique-ID, and infrastructure identifiers were replaced with dummy values.
- Production IP addresses and filesystem paths were replaced with documentation/test values.
- Embedded screenshots containing internal values were reconstructed as synthetic tables/blocks so the document can be safely used for QATOOL testing.
- Business rules, acceptance-criteria wording, section structure, and test-relevant technical patterns were preserved to maintain realistic QA-generation behavior.

End of sanitized test copy.